

TrES-3b

R-1654

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-3_210327 · created 2026-08-02T08:30:39+00:00

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2459300.9338 +/- 0.0014 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.152 +/- 0.019
Transit depth (Rp/Rs)^2	2.31 +/- 0.57 [%]
Orbital Inclination (inc)	83.04 +/- 1.34
Airmass coefficient 1 (a1)	1.0329 +/- 0.0094
Airmass coefficient 2 (a2)	-0.0281 +/- 0.0096
Scatter in the residuals of the lightcurve fit is	0.91 %
Best Comparison Star	#1 - [71, 119]
Optimal Method	PSF photometry
Transit Duration (day)	0.062 +/- 0.011

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.152 ± 0.019 vs 0.16309 (deviation 0.56σ)
Duration fit vs published	0.062 d vs 0.0567 d (deviation 0.47σ)
Mid-time O–C	-4.9 min
Depth SNR	7.7
Residual scatter	0.91 %

Light curve

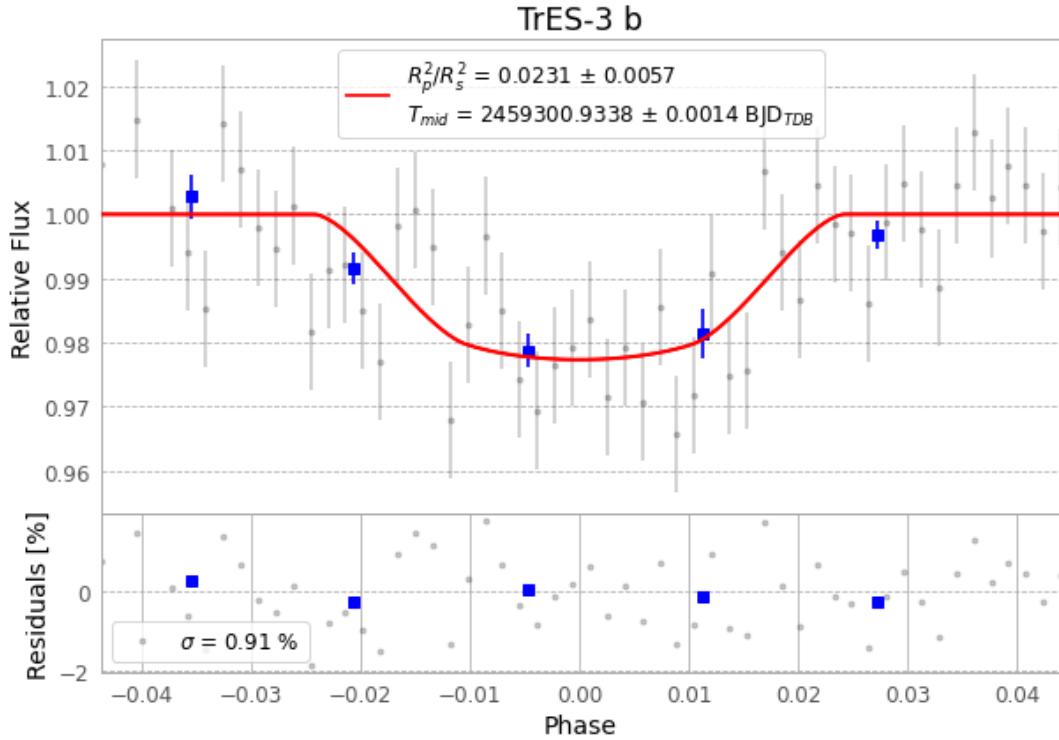


Figure 1 — FinalLightCurve_TrES-3 b_2021-03-27.png (SHA-256 74918209072024d3...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [370, 245], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 74306d3841671ec2...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

55 FITS frames (36 MB), TRES-3210327090010.FITS → TRES-3210327114512.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-3-210327.log (4e5df53cd789...), FinalLightCurve_TrES-3 b_2021-03-27.png (749182090720...), FinalLightCurve_TrES-3 b_2021-03-27.csv (2c963585b41f...)

Compute resources

Wall time 130.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-02 08:30Z.